Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/15/2014
PI Name:	Rubin, David M.S.		
Project Title:	Electronic Medical Records for Exploration Medical System		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **ExMC**:Exploration Medical Capabilities		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77058	**Congressional District:**	36
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	07/01/2011	**End Date:**	03/27/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/27/2015 (from 9/30/2018) per PI (Ed., 6/30/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

The Electronic Medical Record (EMR) is an electronic system that will provide a repository for medical data, images, videos, and reports and make the information accessible to the crew in flight via a graphical user interface to assist in the diagnosis and treatment of acute and chronic medical conditions.
The system will be used to access medical data for all medical encounters, especially during periods of exploration missions when contact with ground resources (e.g. flight surgeon) will either be minimal or absent. During these periods of autonomous medical care, the crew will need such a system for urgent and non-urgent medical events.

Development of the EMR addresses the following risk as stated within the Critical Path Roadmap (CPR; also known as the NASA Bioastronautics Roadmap (NASA/SP-2004-6113)):

Risk of Unacceptable Health and Mission Outcomes Due to Limitations of In-flight Medical Capabilities: Given that

Task Description:	medical capabilities will be limited during human exploration missions, there is a possibility that in-flight medical events will lead to unacceptable health and mission outcomes. The objectives of the EMR are to: a. Reduce or eliminate the time required of an on-orbit crew (i.e. clinician and non-clinician) to store, access, and process medical data, images, and reports. b. Reduce the time required of the ground crew to access medical data, images, or reports. c. Ensure CMO access to all relevant crewmember medical data, images, or reports for improved quality of care. d. Ensure crew access to personal medical data, images, or reports during periods of restricted ground communication. e. Ensure crew ability to capture onboard medical data regardless of the state of the communication link to the ground. f. Ensure privacy protection of onboard crewmember medical data. The ground demonstration of the EMR, as part of the Exploration Medical System Demonstration (EMSD), will occur in 2015 and the in-flight phase will take place onboard the ISS in 2018.
Rationale for HRP Directed Research:	This study is a combination of a solicited task (RFP released under Wyle for procurement of COTS EMR system) and a directed task involving customization of COTS software for the specific needs of ExMC. The study team is uniquely positioned to perform this function and to ensure that the EMR component of the overall Exploration Medical System Demonstration (EMSD) is integrated with the other components of the system.
Research Impact/Earth Benefits:	The Electronic Medical Record, as part of the Exploration Medical System, is being designed specifically for a Near Earth Asteroid mission and, as such, has no direct impact or benefits to life on Earth. However, development of the OpenEMR application programming interfaces (APIs), required for integration into the Exploration Medical System, can be made available to the OpenEMR open source community for potential inclusion into future releases of the EMR software. In addition, the Exploration Medical System can serve as software infrastructure for future development and integration of medical devices and software.
Task Progress:	The following accomplishments have been made during the current reporting period of 6/30/13-7/1/14: The EMSD team dispositioned and implemented all RIDs from Preliminary Design Review process. EMSD project documents were updated and an Authority to Proceed (ATP) was received from the Science Management Office (SMO). -The EMSD team participated in the ExMC System Requirements Review (Phase 3) by submitted updates to the ExMC Operational Concepts (OpsCon) document and Functional Requirements Document (FRD), including a new section for the image and video capture system of EMSD. The EMSD team also participated in the RID review and disposition process for these documents. -The EMSD team procured and installed the EMR software on the test computing platform. The project initially procured the commercially available EMR software but, after further evaluation, the team determined that required development was cost prohibitive. The project then switched to the open source OpenEMR software. The team is currently developing additional functionality for the OpenEMR software to support full integration of the EMSD system. -The EMSD team conducted the Detailed Design Phase of the project. As part of this phase, the EMSD team conducted the Critical Design Review Kick Off and RID Board meetings. All RIDs were dispositioned and implemented and all project documents were updated accordingly. There were no impacts to the EMR or integration of the EMR into the EMSD system. -The EMSD team developed a Validation and Verification Document which will be used, in part, to validate and verify EMR functionality within the EMSD system. -The EMSD team began the Production, Test, and Certification Phase of the EMSD project. This includes the development of additional EMR functionality (e.g. APIs, visit templates), testing of that functionality, and integration of the EMR with the other EMSD components. -The EMSD team integrated the NSBRI SpaceMED system into the EMSD system. This includes the transmittal of data, produced by the SpaceMED system, into the EMSD EMR software.
Bibliography Type:	Description: (Last Updated: 04/16/2014)
Abstracts for Journals and Proceedings	Rubin DA, Watkins SD. "Exploration Medical System Demonstration Project." Presented at the 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3047.pdf , Feb-2014
Abstracts for Journals and Proceedings	Shah RV, Rubin DA, Kerstman EL. "NASA Exploration Medical Capability Exploration Medical System Demonstration Project." Presented at the 85th Annual Scientific Meeting, Aerospace Medical Association, San Diego, CA, May 10-15, 2014. Aviation, Space, and Environmental Medicine. 2014 Mar;85(3):327. See http://www.ingentaconnect.com/content/asma/asem/2014/00000085/00000003 , Mar-2014